Evan Lowry

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Education

University of Waterloo Waterloo, ON

Honours BMath in Computational Mathematics, Co-op *Sep. 2025 - Aug. 2030*

Major Average: 84%

Relevant Coursework: Object-Oriented Software Development, Logic and Computation, Probability

Clubs/Activities: Quant Club, Data Science Club, WAT.ai, UW Blueprint

Experience

Full Stack Developer & Project Manager Apr. 2025 – Sep. 2026

Pizzeria La Terrazza *Kitchener, ON*

- Launched a pizzeria's first online ordering site on **Square**, creating a new sales channel and processing **$40,000+** in online orders, increasing average site traffic **3x** to **900+** unique monthly visitors in its first year deployed.
- Cold-pitched project to the client, managed scoping, requirements, follow-ups, onboarding and ongoing support.
- Rebuilt the platform in **Next.js**, **React**, and **TypeScript** with **Square** payments, automated emails and a **CMS**.
- Added **Cloudflare** analytics to track page traffic and latency, used data to drive changes to the website layout and content, increasing the conversion rate by over **15%**.

Full Stack Developer Apr. 2026 – Present

UW Blueprint, Don't Mess With the Don *Waterloo, ON*

- Developing an event management platform for a local nonprofit, designed to support **300+ annual events** and **8,000+ attendees**, collaborating with six developers and six designers and participating in client meetings.
- Built an asynchronous attendance API using **FastAPI, SQLAlchemy, and PostgreSQL**, enforcing unique check-ins and enabling a streamlined flow for volunteers and attendees, expected to **save hours** per event.
- Implemented a **timezone-aware CRON job** system supporting multiple synchronous and asynchronous jobs.
- Developed a reusable **email and SMS service** for volunteer outreach, integrating with scheduled reminders.

Research Contributor Aug. 2026 – Present

University of Waterloo *Waterloo, ON*

- Contributing to open-source search and **LLM reranking** tools, improving documentation and CLI workflows.
- Running experiments on cloud **GPUs**, assessing retrieval quality, combining classical search with neural rerankers.

Projects

Derive AI | React, TypeScript, Express, MongoDB Feb. 2026

- Used **OpenRouter** to integrate LLMs that read and respond to problems interactively, graph and verify results.
- Implemented **Auth0**-protected notes and chat APIs with bulk sync, per-user indexed storage, and isolation tests.
- Worked in a three-person team to build a canvas whiteboard with stroke stabilization, multi-object selection and transforms, shape tools, pan/zoom, and undo/redo functionality.

Rubik's Cube Solver | Java Dec. 2025 – Jan. 2026

- Engineered a two-phase cube solver using **iterative-deepening search** exploring an average of **2 billion states**.
- Developed a precomputed heuristic database and move-sequence pruning to eliminate redundant branches and reject states that are unable to meet the target state within the remaining search depth, averaging **26** moves per solve.

3D Rendering Engine | Java, Swing, Java2D Nov. 2024 – Jan. 2025

- Created a **software rendering engine**, with camera transforms, perspective projection, and textured triangles.
- Adopted perspective-correct **UV interpolation**, near-plane clipping, backface culling, and normal-based lighting.
- Developed an **OBJ asset-loading** pipeline for geometry, normals, UV coordinates, and material assignments; integrated mouse-look controls, gravity, jumping, and wall collision detection.

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, Racket
Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, SQLAlchemy
Databases: PostgreSQL, MongoDB
Testing: Jest, Vitest, Playwright, Pytest, JUnit
Developer Tools: Git, GitHub Actions, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ